

R Notebook

This is an R Markdown Notebook. When you execute code within the notebook, the results appear beneath the code.

Try executing this chunk by clicking the *Run* button within the chunk or by placing your cursor inside it and pressing *Cmd+Shift+Enter*.

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(tibble)
library(lubridate)
```

```
##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##   date, intersect, setdiff, union
```

```
library(ggplot2)
library(scales)
```

```
library(readxl)
sportsref_download_6_ <- read_excel("Desktop/sportsref_download (6).xlsx")
AngelsAttendance <- sportsref_download_6_
#View(sportsref_download_6_)
```

```
names(AngelsAttendance)
```

```
## [1] "Year"      "Tm"        "Lg"        "W"         "L"
## [6] "Finish"    "Playoffs"  "Attendance" "Attend/G"  "Rank"
## [11] "Est. Payroll" "PPF"      "BPF"      "Stadium"
```

First I will create a new variable that shows the winning percentage:

```
AngelsAttendance <- AngelsAttendance %>%
  mutate(WinPerc = round(W / (W + L), 2))
```

Replace NAs in Playoffs with “Missed Playoffs”

```
AngelsAttendance$Playoffs[is.na(AngelsAttendance$Playoffs)] <- "Missed Playoffs"
```

We already know there was no attendance in 2020 due to the COVID-19 pandemic so we will ignore that season and also 2021 since a full crowd was not present until 2022.

2022-2025

```
AngelsAtt2022to2025 <- AngelsAttendance %>% filter(Year %in% (2022:2025))
```

Every season after the 2014 playoff appearance

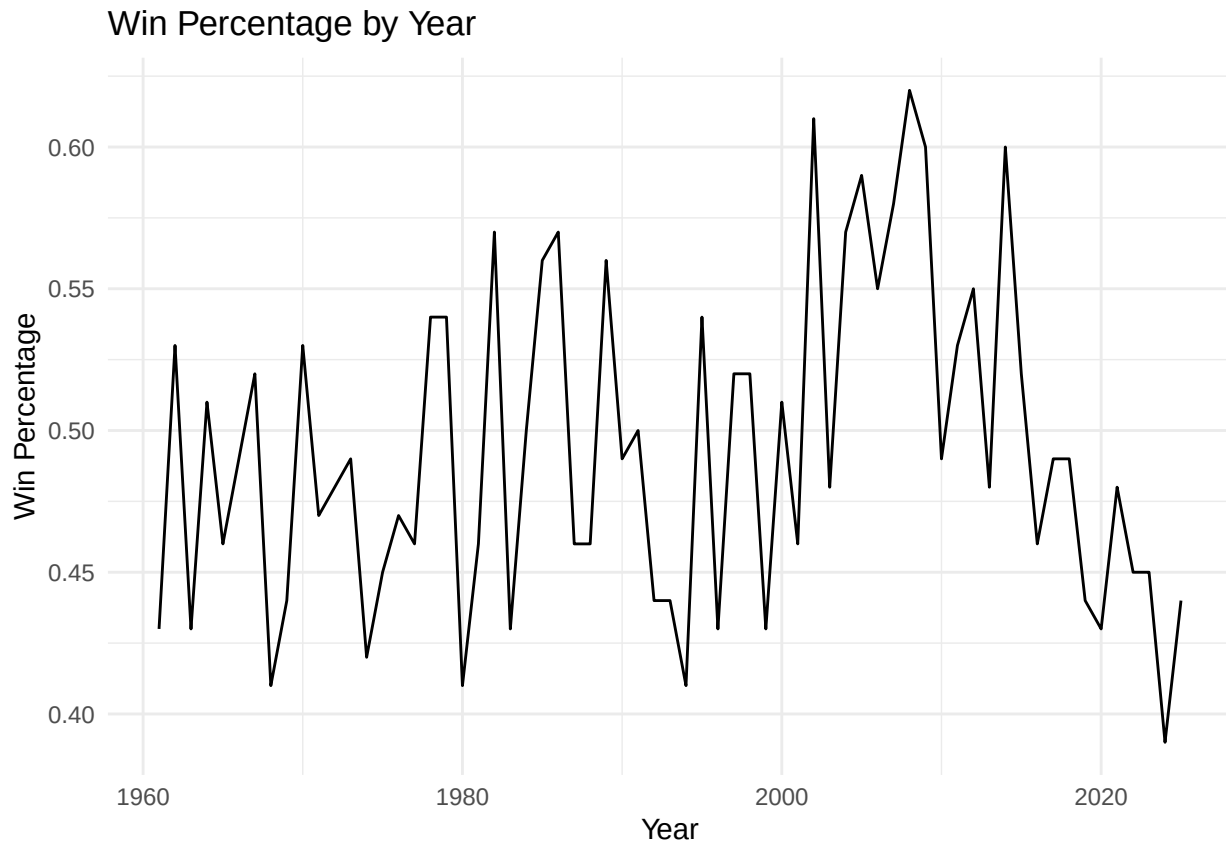
```
LAAattafterplayoffs <- AngelsAttendance %>% filter(Year >= 2014)
```

Excluding the COVID seasons

```
AngelsAttNOCOVID <- AngelsAttendance %>% filter(!Year %in% c(2020, 2021))  
LAAAttPlayoffsNoCOVID <- LAAattafterplayoffs %>% filter(!Year %in% c(2020, 2021))
```

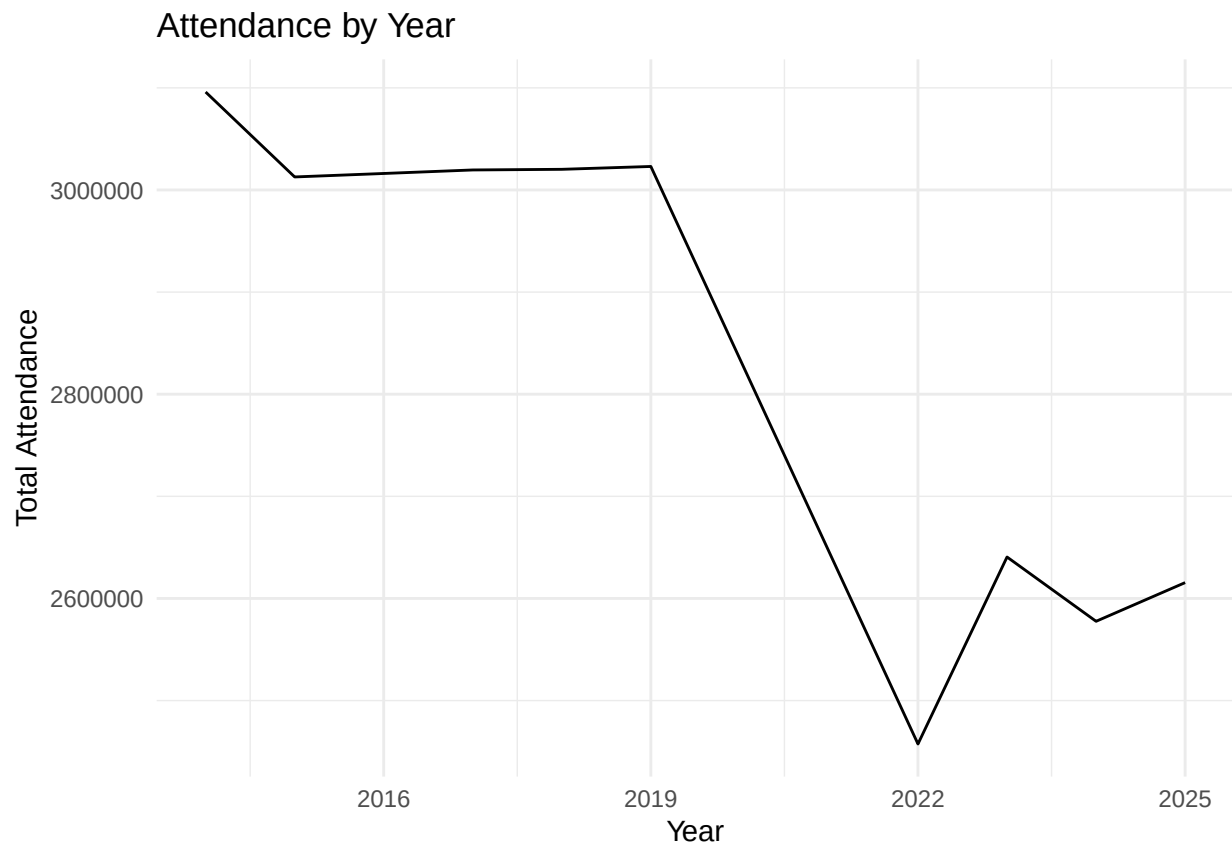
This line graph shows that the Angels are currently performing their franchise worst.

```
ggplot(AngelsAttendance, aes(x = Year, y = WinPerc)) +  
  geom_line() +  
  labs(title = "Win Percentage by Year", x = "Year", y = "Win Percentage") +  
  theme_minimal()
```

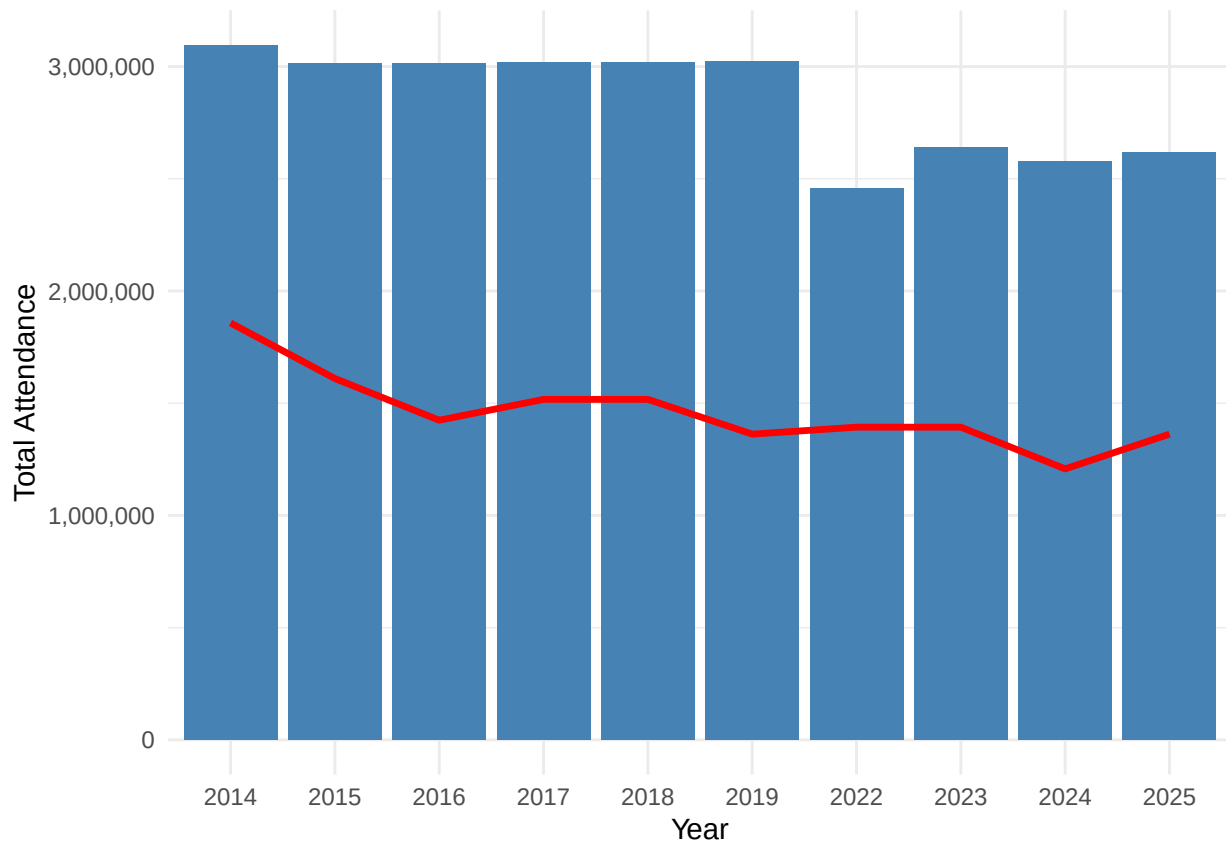


I will use the data excluding 2020 and 2021 since attendance figures were low during those years:

```
ggplot(LAAAttPlayoffsNoCOVID, aes(x = Year, y = Attendance)) +  
  geom_line() +  
  labs(title = "Attendance by Year", x = "Year", y = "Total Attendance") +  
  theme_minimal()
```



```
ggplot(LAAAttPlayoffsNoCOVID, aes(x = factor(Year), y = Attendance)) +  
  geom_bar(stat = "identity", fill = "steelblue") +  
  geom_line(aes(y = WinPerc*max(Attendance), group = 1), color = "red", linewidth = 1.2) +  
  labs(x = "Year", y = "Total Attendance") +  
  scale_y_continuous(labels = comma) +  
  theme_minimal()
```

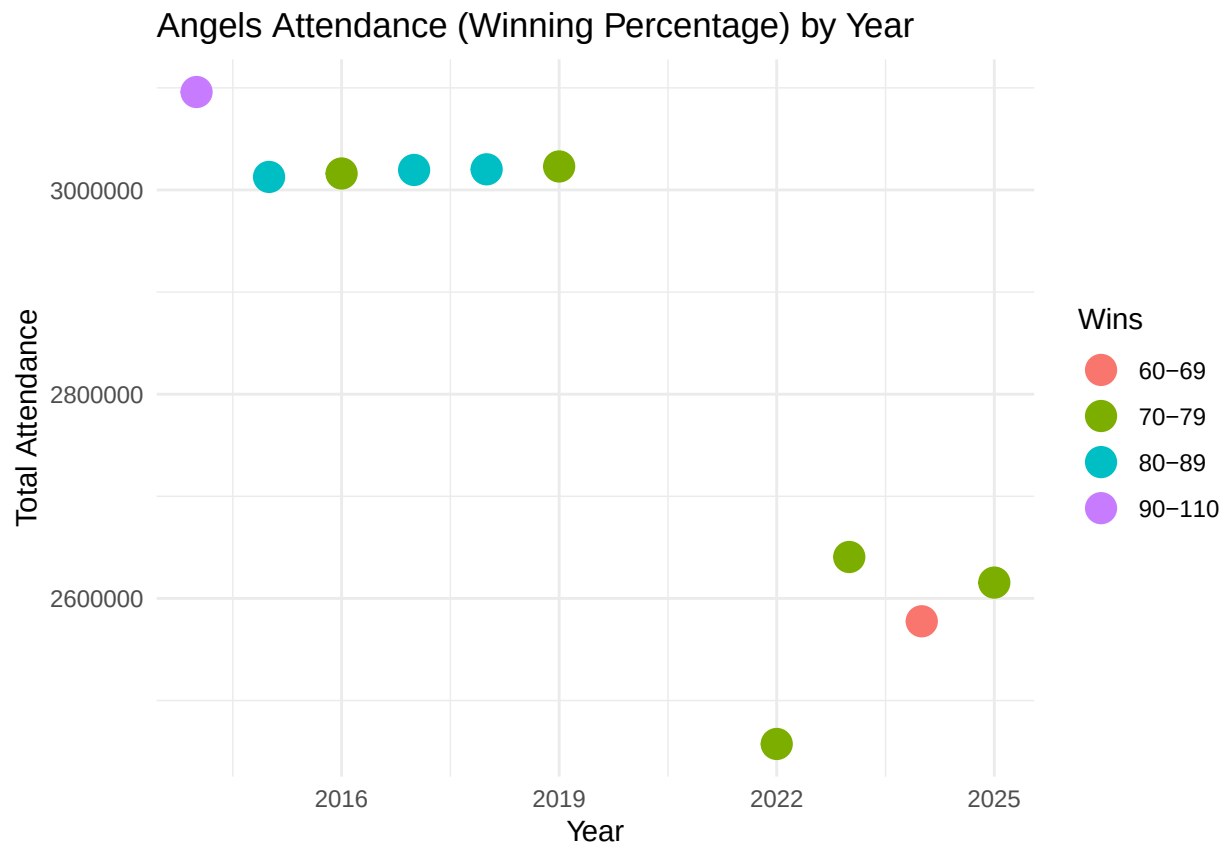


```
LAAAttPlayoffsNoCOVID$win_group <- cut(
  LAAAttPlayoffsNoCOVID$W,
  breaks = c(60, 70, 80, 90, 110),
  labels = c("60-69", "70-79", "80-89", "90-110"),
  right = FALSE
)
```

```
AngelsAttNOCOVID$win_group <- cut(
  AngelsAttNOCOVID$W,
  breaks = c(60, 70, 80, 90, 110),
  labels = c("60-69", "70-79", "80-89", "90-110"),
  right = FALSE
)
```

```
ggplot(LAAAttPlayoffsNoCOVID, aes(x = Year, y = Attendance, color = win_group)) +
  geom_point(size = 5) +
  scale_fill_manual(values = c(
    "60-69" = "black",
    "70-79" = "blue",
    "80-89" = "green",
    "90-110" = "red")) +
  labs(x = "Year", y = "Total Attendance", color = "Wins", title = "Angels Attendance (Winning Percentage)",
  theme_minimal()
```

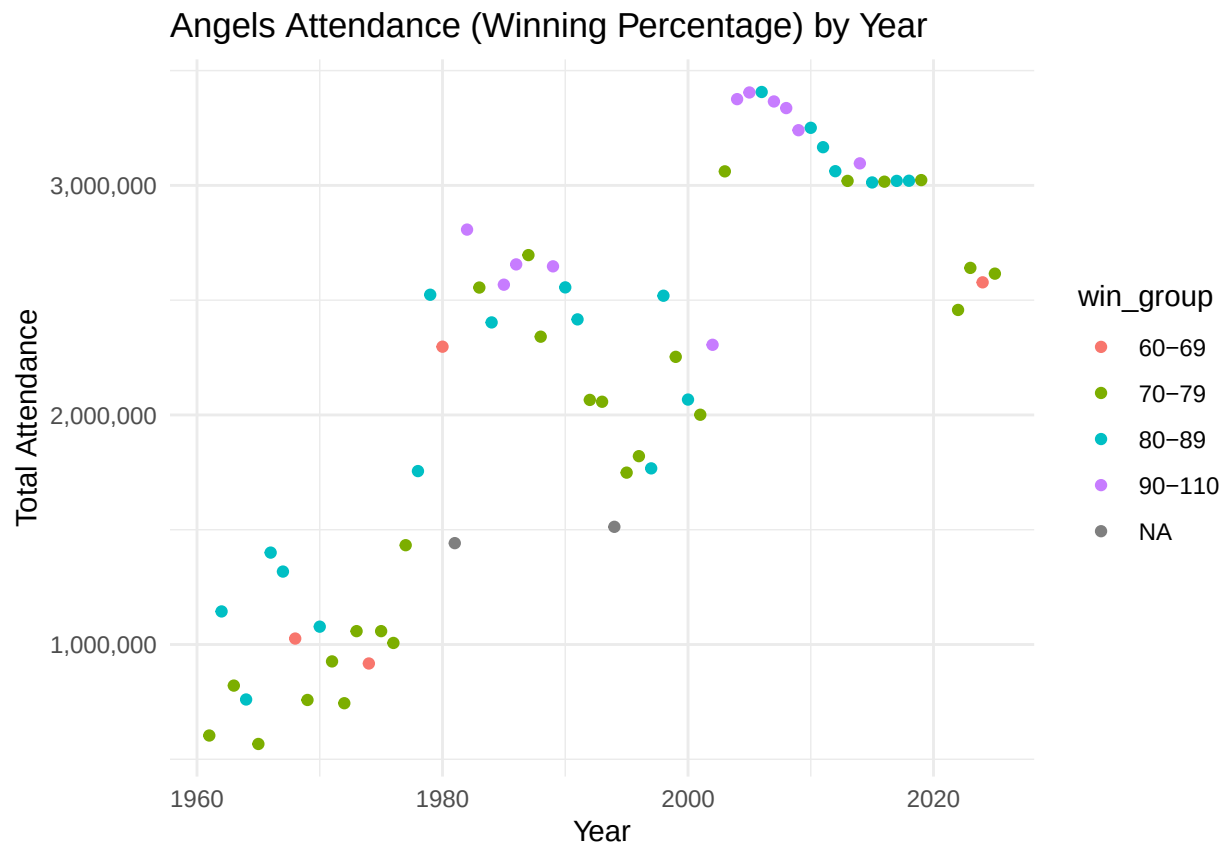
```
## Warning: No shared levels found between `names(values)` of the manual scale and the
## data's fill values.
```



Angels have never lost 100 games. The NAs are the 1981 and 1994 seasons, which were shortened.

```
ggplot(AngelsAttNOCVID, aes(x = Year, y = Attendance, color = win_group)) +
  geom_point() +
  scale_fill_manual(values = c(
    "60-69" = "black",
    "70-79" = "blue",
    "80-89" = "green",
    "90-110" = "red")) +
  labs(x = "Year", y = "Total Attendance", title = "Angels Attendance (Winning Percentage) by Year") +
  scale_y_continuous(labels = comma) +
  theme_minimal()
```

```
## Warning: No shared levels found between `names(values)` of the manual scale and the
## data's fill values.
```



On the side, I wanted to see how many wins the Angels were projected to win at the time I did this side project

July 10, 2026: Record 36-56 Runs Scored: 399 Runs Allowed: 457

Pythagorean Expectation: $W\% = \frac{(\text{Runs Scored})^{1.81}}{(\text{Runs Scored})^{1.81} + (\text{Runs Allowed})^{1.81}}$

```
PE = (399^1.81)/(399^1.81+457^1.81)
```

```
PE*162
```

```
## [1] 71.10063
```

```
projected_wins <- current_wins + (remaining_games * current_win_pct)
```

```
ProjectedWins <- 36 + (70 * .391)
```

```
ProjectedWins
```

```
## [1] 63.37
```

```
36 + 70*PE
```

```
## [1] 66.7225
```